

Egocentric Videos Understanding with Active Memory

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Si può costruire una memoria attiva
semantic-free a partire
da un video egocentrico?

Si può costruire una memoria attiva
semantic-free a partire
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Si può costruire una memoria attiva
semantic-free a partire
da un video egocentrico?



Si può costruire una memoria attiva semantic-free a partire da un video egocentrico?

Egocentric Camera



View From Egocentric Camera



Use cases

Use cases

Vita quotidiana

Industria

Sicurezza

Problemi

Problemi

Lunghi

Frammentati

Rumore

Accuracy

AMEGO

AMEGO

Stato dell'arte

Semantic-free

AMB



Struttura

Struttura

Location segments

HOI tracklets

Struttura

Location segments

HOI tracklets



Struttura

Location segments

HOI tracklets



HOI tracklets /elementi

Timestamp interazione

Bounding box oggetto

Mano coinvolta

HOI tracklets /fasi

Init

Update

Store

HOI tracklets /elementi

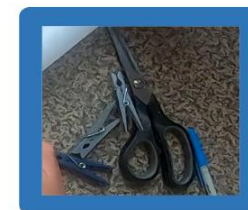
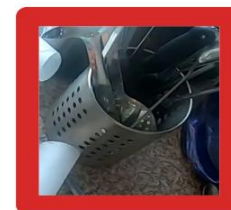
Init

Initialisation

New
tracklet o_i

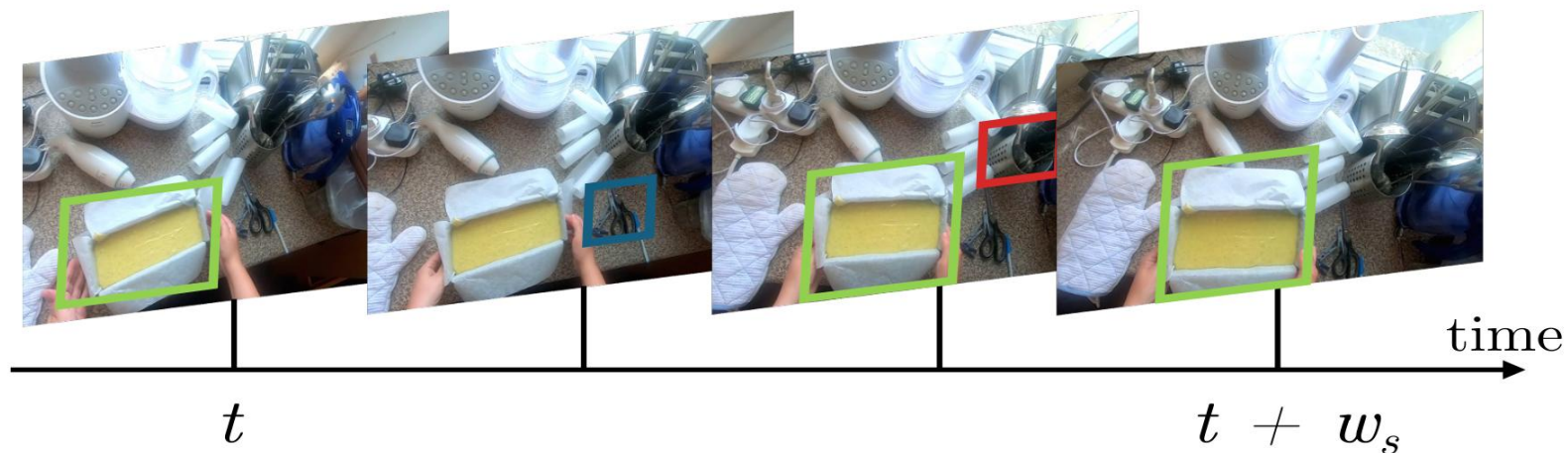


Reject



Update

Store

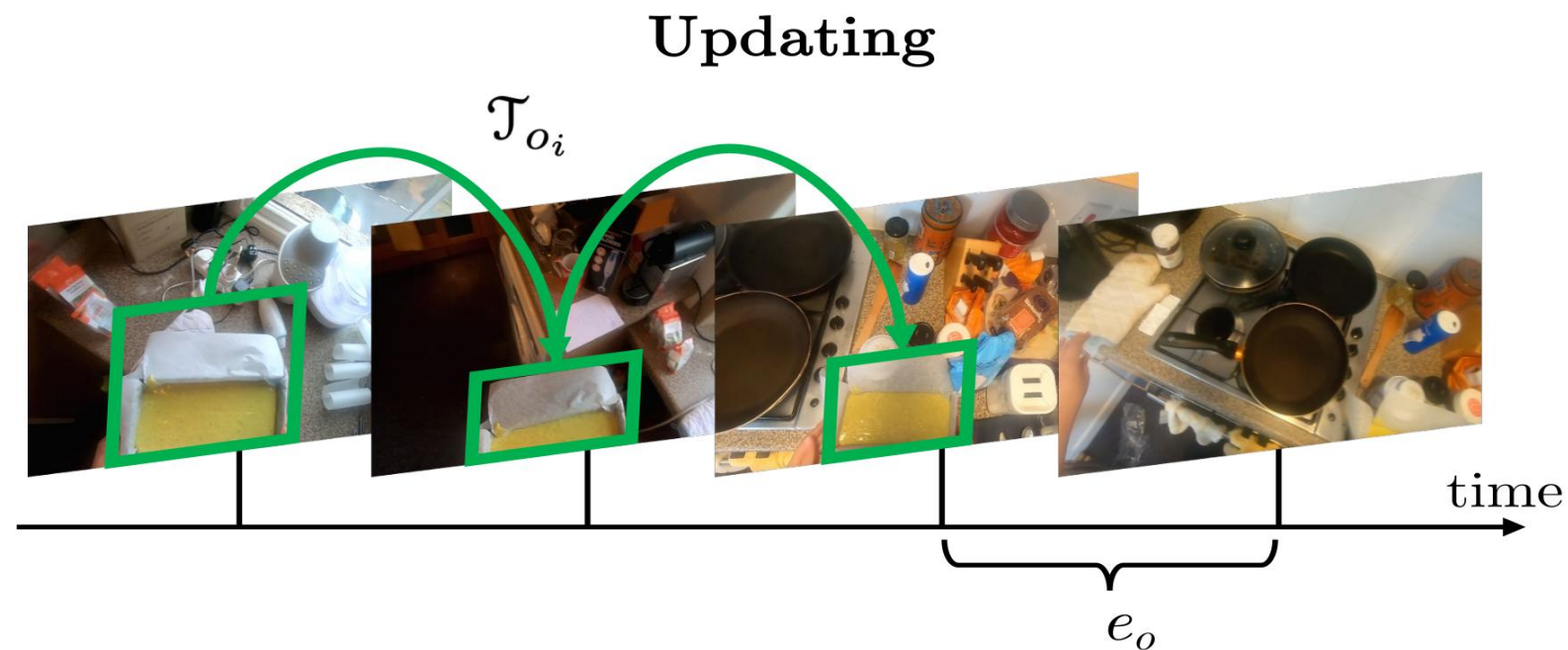


HOI tracklets /elementi

Init

Update

Store

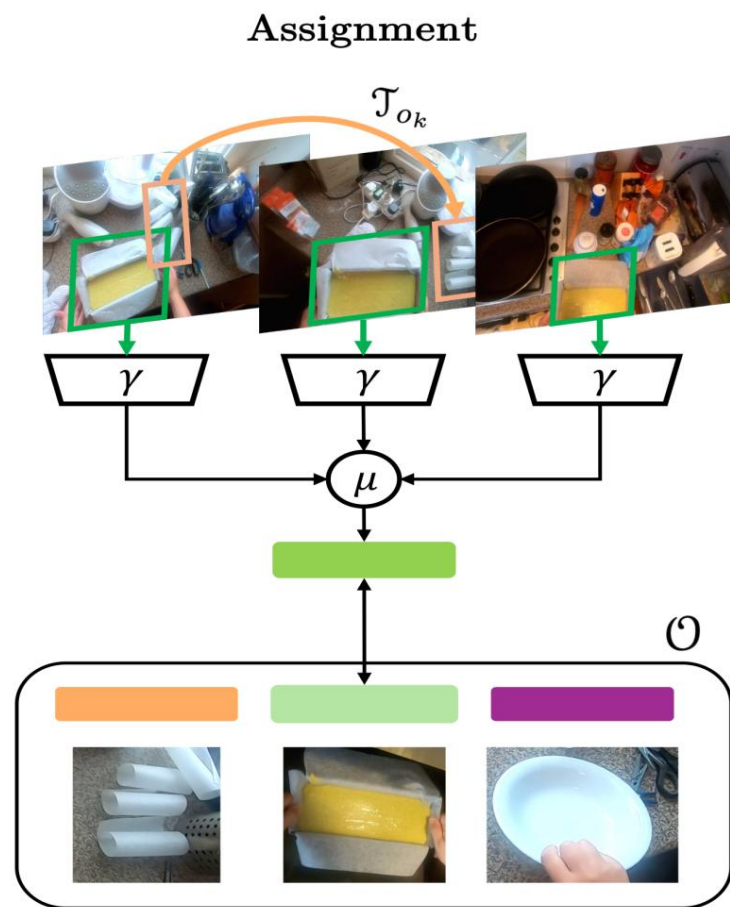


HOI tracklets /elementi

Init

Update

Store



Esperimenti

Valutare la capacità di
generalizzare in contesti
complessi

Esperimenti /obiettivo

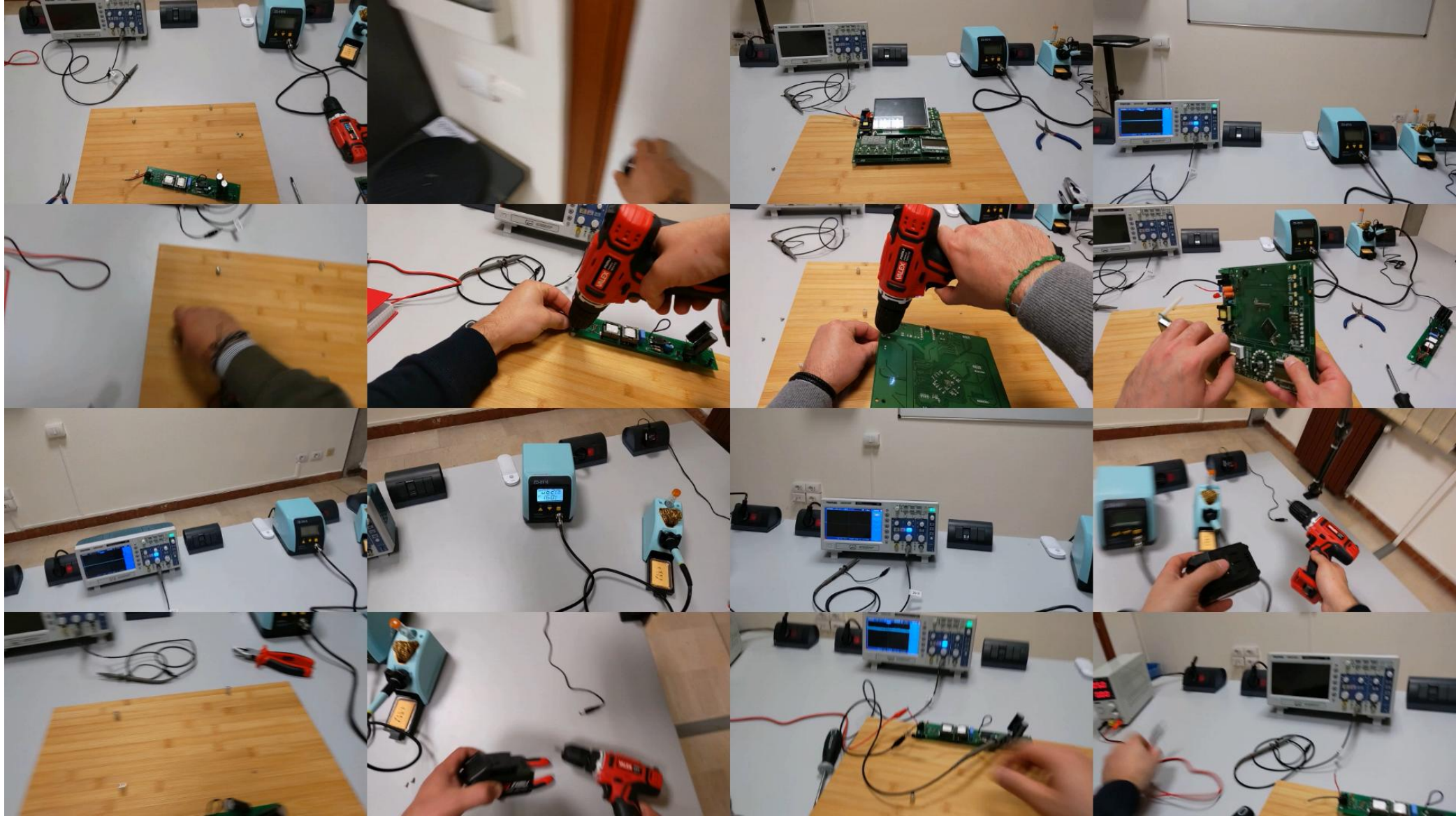
“Con quali oggetti ho interagito
simultaneamente?”

Esperimenti /query di valutazione



Esperimenti /datasets

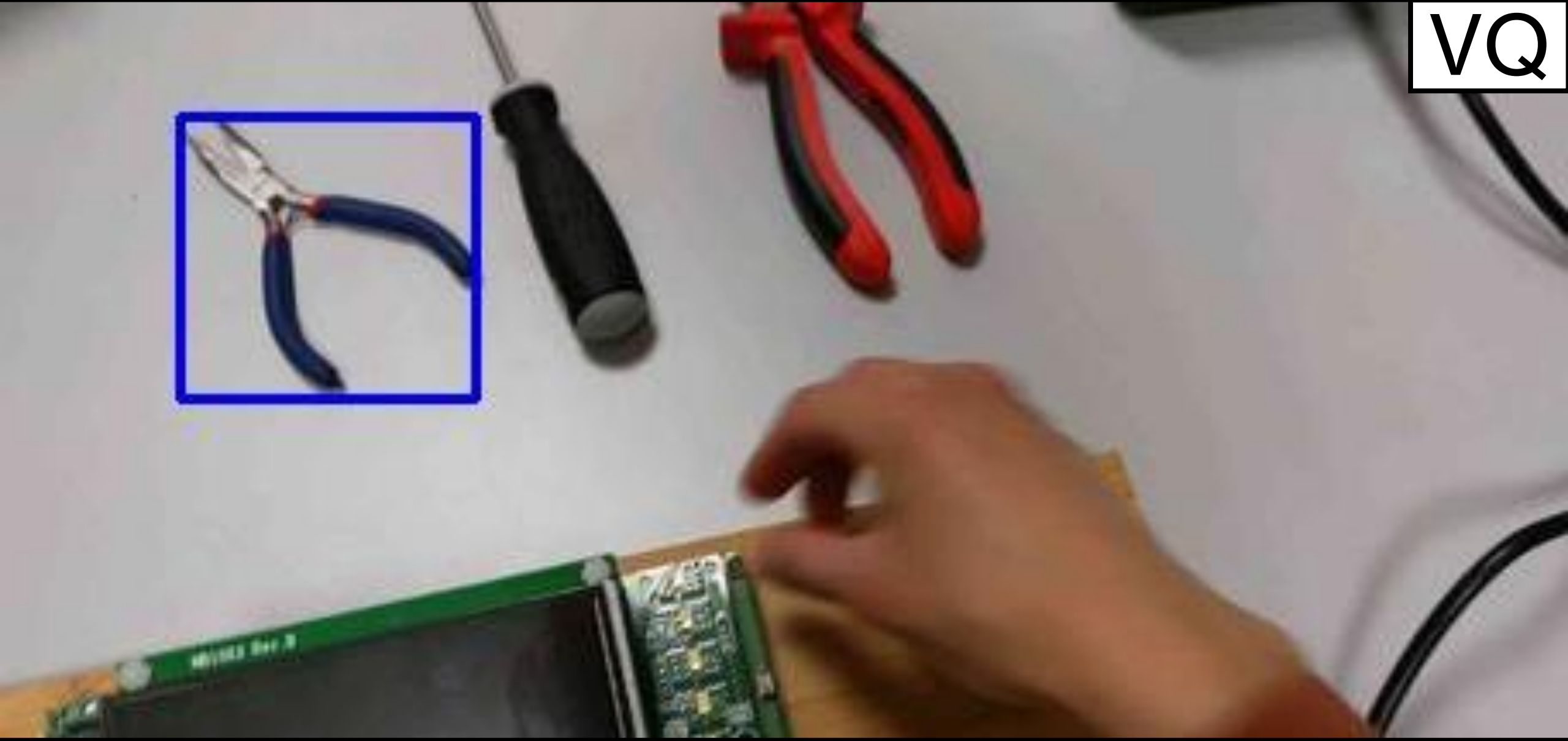
EPIC KITCHEN



Esperimenti /datasets

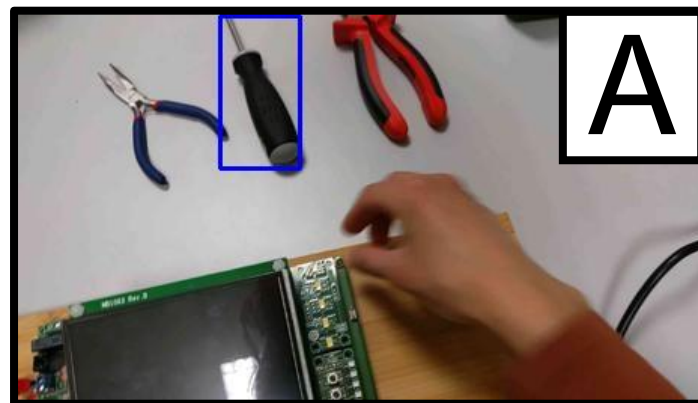
Q5_0000388

VQ

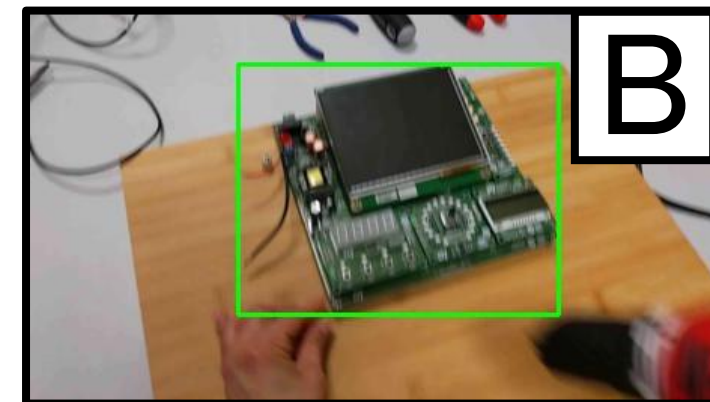


Q5_000388

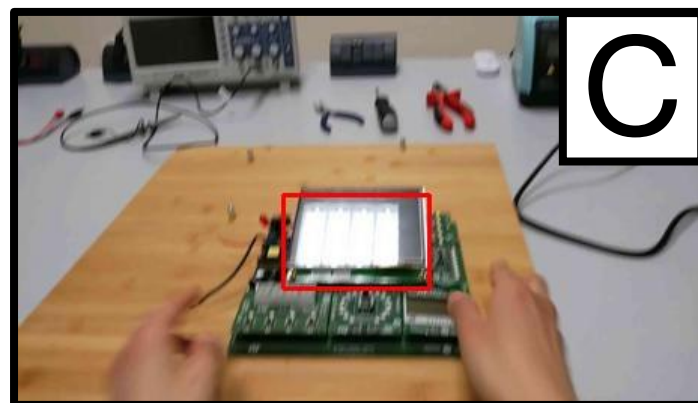
VQ



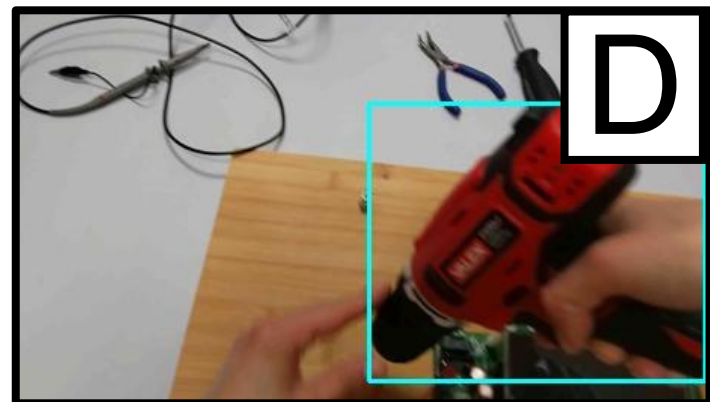
A



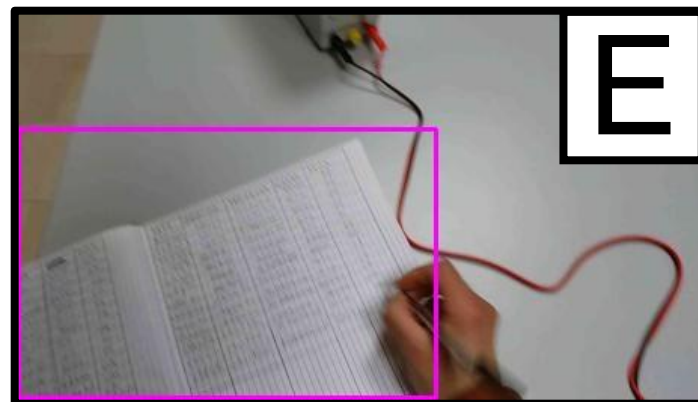
B



C



D

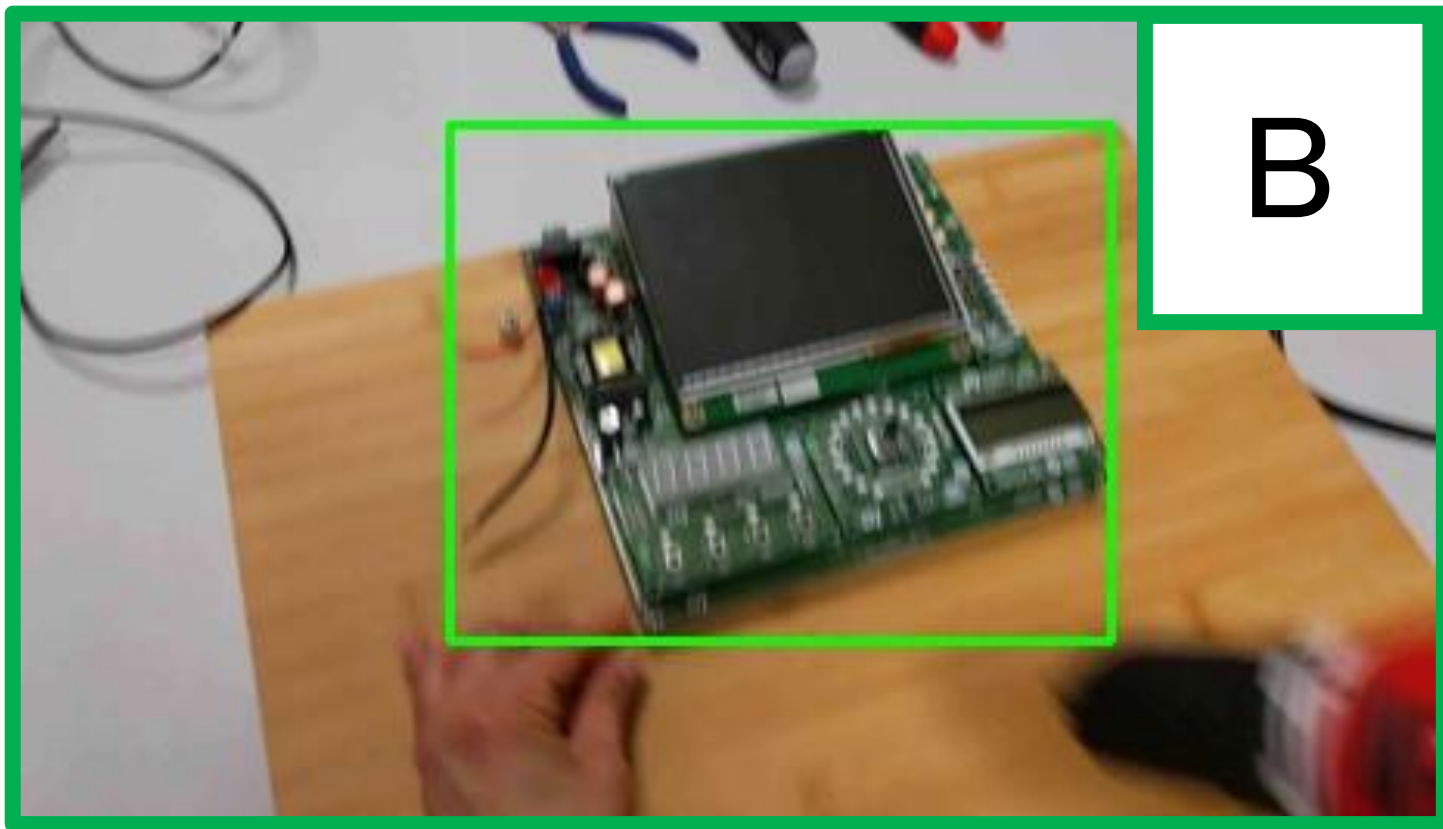


E

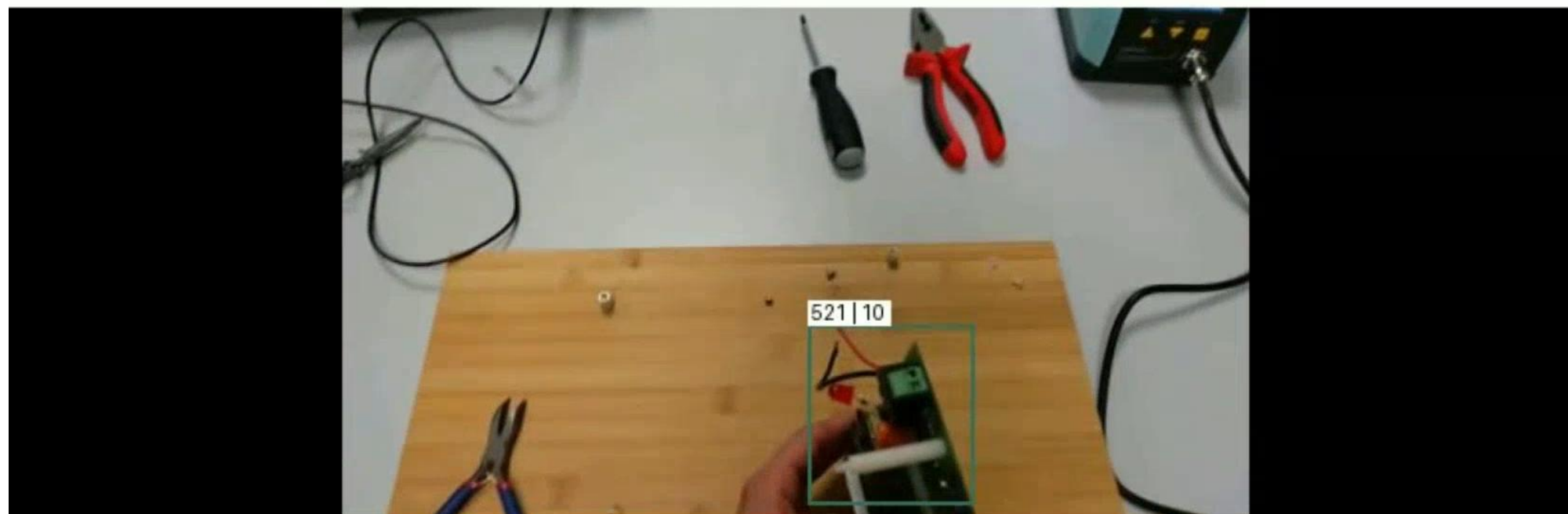
VQ



B



video_id: 89 frame 15054/51444



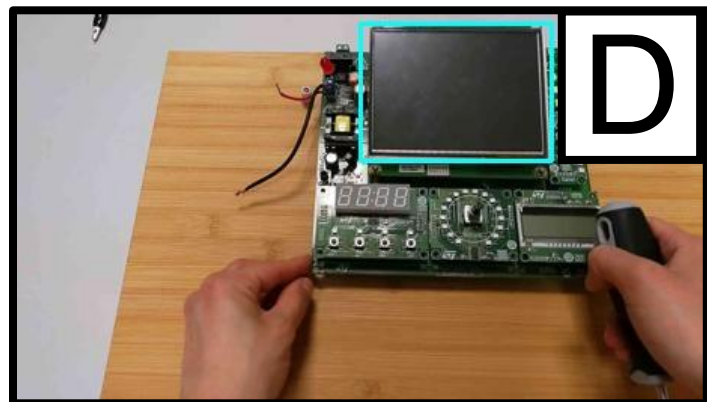
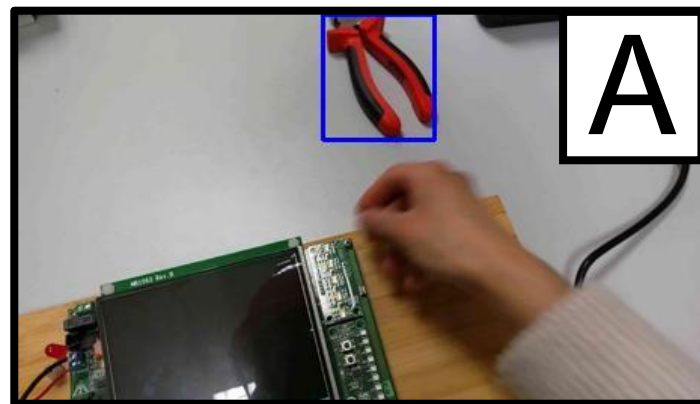
cluster	track_id	color	time_left_active	time_to_be_active
10	521		94	0
15	525		0	32
15	530		0	58
15	532		0	80
2	552		0	530

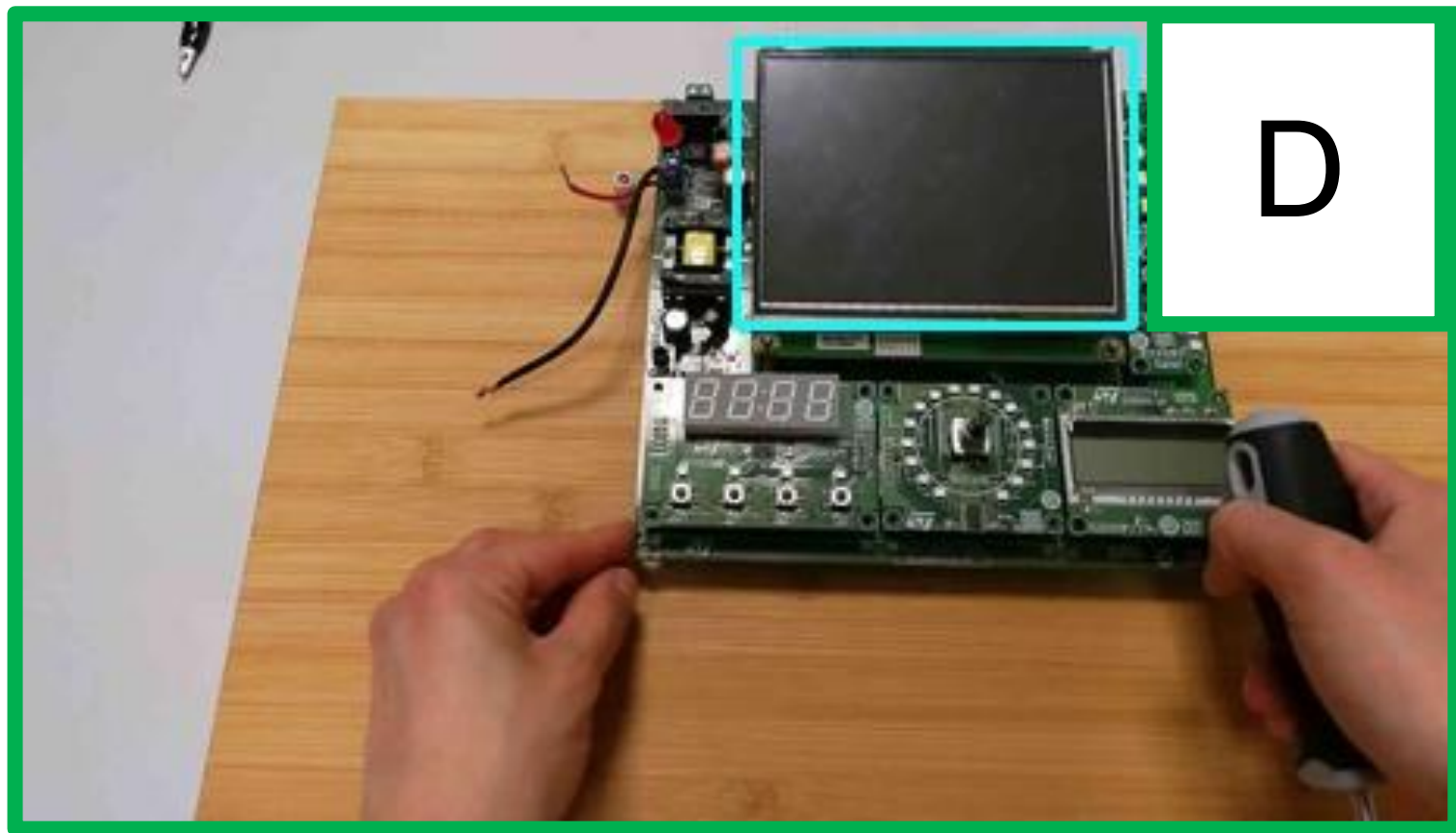
Q5_0000104

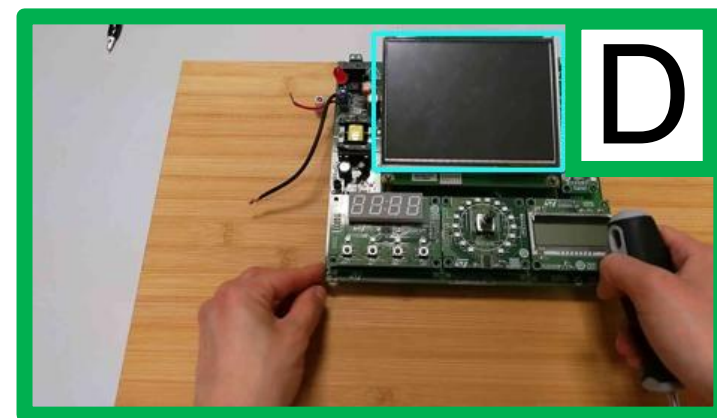
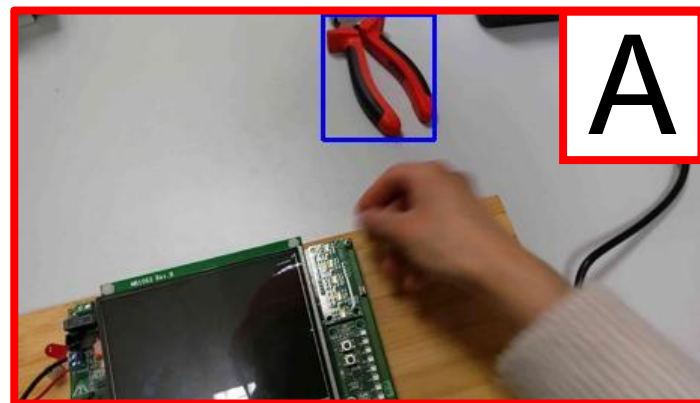
VQ



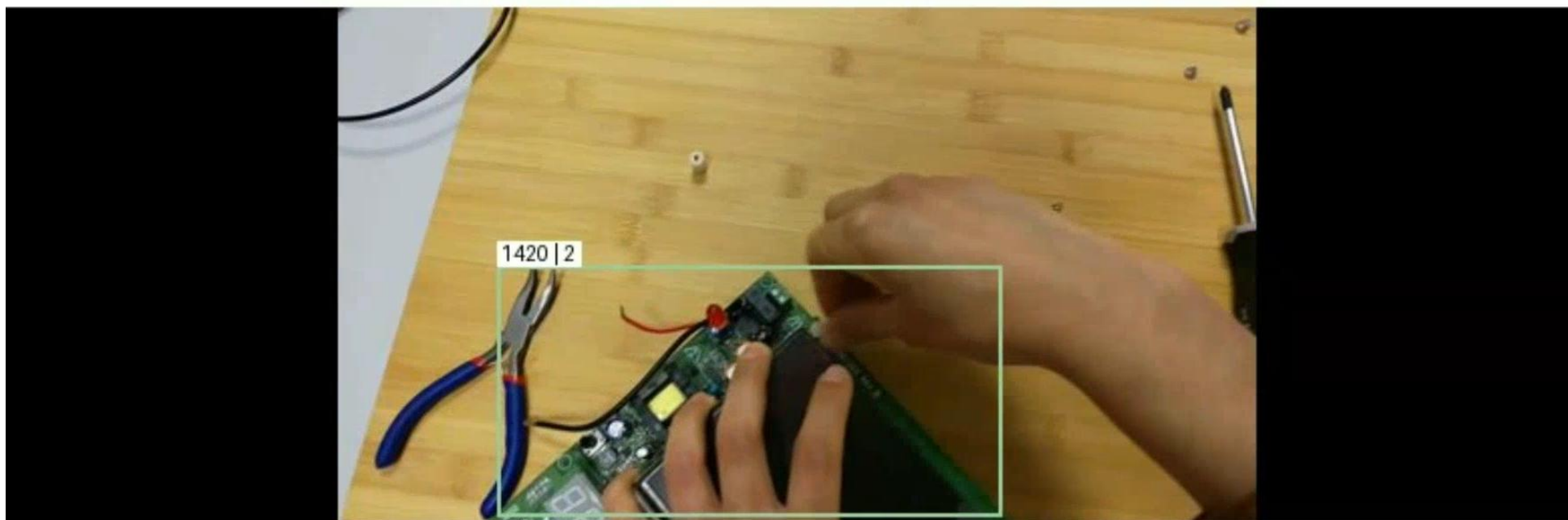
Q5_000104







video_id: 53 frame 30424/50203



cluster	track_id	color	time_left_active	time_to_be_active
2	1420		418	0
9	1439		0	442
12	1442		0	500
12	1454		0	647
12	1457		0	665

Risultati

EPIC KITCHEN

27.6%

Risultati

ACCURACY

ENIGMA-51

23.7%

Risultati

ACCURACY

ENIGMA-51

28.9%

~~23.7%~~

Risultati

ACCURACY RELAXED

Conclusioni

Conclusioni

Limiti

Precisione

Euristica

Possibili soluzioni

Sviluppi futuri

Conclusioni

Limiti

Fine-tuning

Possibili soluzioni

Risoluzioni
frame

Sviluppi futuri

Conclusioni

Limiti

Possibili soluzioni

Cambio
euristiche

Sviluppi futuri

Query

Dataset

Grazie